Ariel Szekely

Contact Information

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Education

2020-present **Ph.D. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA

2020-2022 **M.S. in Computer Science**, *Massachusetts Institute of Technology*, Cambridge, MA,
Thesis: *σOS: Elastic Realms for Multi-Tenant Cloud Computing*
GPA: 5.0/5.0

2016-2020 **B.S. in Computer Science (Honors)**, *The University of Texas at Austin*, Austin, TX,
Thesis: *Exploring GPU Timing Side-Channels*
GPA: 3.94/4.0

Interests

Distributed Systems, Operating Systems, Security, Concurrency.

Professional Experience

2023 **Student Researcher**, *Google Systems Research Group*, Seattle, WA
Designed and evaluated a system for TPU disaggregation for Ads model serving.

2019 **Research Assistant**, *The University of Texas at Austin*, Austin, TX
Developed the Telekine GPU timing side-channel attack for our NSDI '20 paper.

2018 **Software Engineer Intern**, *Bloomberg L.P.*, New York City, NY
Back-end development.

2017 **Software Engineer Intern**, *AT&T*, Dallas, TX
Full-stack development.

Publications

Eurosys '27 (under submission) **Fast end-to-end cloud application cold-start with co-sandboxes**, **Ariel Szekely**, *Hannah Gross, Robert Morris, Frans Kaashoek*
Designed and implemented co-sandboxes, a new scriptable interface for cloud developers to specify their application's initialization routine to the provider in order to overlap application initialization with infrastructure setup costs. Co-sandboxes reduce cold-start latency of serverless applications by 1.6-1.8X and microservices like etcd and memcached by 1.7X.

SOSP '24 **Unifying serverless and microservice workloads with SigmaOS**, **Ariel Szekely**, *Adam Belay, Robert Morris, M. Frans Kaashoek*
Designed and implemented SigmaOS, a novel cloud operating system and cluster manager which supports both microservices and serverless applications. SigmaOS's API enables applications to warm-start in under 2ms and cold-start in under 8ms. [GitHub]

ASPLOS '23 **Towards a machine learning-assisted kernel with LAKE**, *Henrique Fingler, Isha Tarte, Hangchen Yu,* **Ariel Szekely**, *Bodun Hu, Aditya Akella, Christopher J. Rossbach*
Rewrote the Linux Kernel Samepage Merging (KSM) module to benefit from hardware acceleration.

NSDI '20 **Telekine: Secure computing with cloud GPUs**, *Tyler Hunt, Zhipeng Jia, Vance Miller,* **Ariel Szekely**, *Yige Hu, Christopher J. Rossbach, Emmett Witchel*
Built a timing side-channel attack against a GPU Trusted Execution Environment (TEE), allowing the attacker to learn the class of images passed through ResNet by timing the GPU kernels.

Awards

2022 NSF Graduate Research Fellowship
2020 MIT Lemelson Presidential Fellowship
2020 MIT EECS Departmental Fellowship
2020 Dean's Honored Graduate, UT Austin College of Natural Sciences
2020 Research Distinction, UT Austin College of Natural Sciences
2020 NSF Graduate Research Fellowship Honorable Mention
2020 CRA Outstanding Undergraduate Researcher Honorable Mention
2019 NSF REU
2019 UT Endowed Presidential Scholarship
2018,19,20 UT College of Natural Sciences College Scholar
2018 Louis E. Rosier Memorial Scholarship
2016 UT CNS Freshman Scholarship
2016 National Merit Scholarship

Mentorship and Advising

2026 **Nicolas Camenisch**, *Fork Around and Find Out: Accelerating Dynamic Language Runtimes in SigmaOS via Lightweight Container Forking*, M.Eng. Thesis

2026 **Nathan Mustafa**, *Efficient RDMA for SigmaOS*, M.Eng. Thesis

2025 **Ivy Wu**, *Interposing the syscall boundary: Transparent python execution in SigmaOS*, M.Eng. Thesis

2025 **Ryan Chang**, *Optimizing SigmaOS for efficient orchestration of fault-tolerant, burst-parallel workloads*, M.Eng. Thesis

2025 **Freddie Tang**, *Accelerating Burst Parallelism of SigmaOS processes with CRIU*, M.Eng. Thesis

2025 **Katie Liu**, *Enabling efficient ML inference in SigmaOS with model-aware scheduling*, M.Eng. Thesis

2023 **Yizheng He**, *Evaluating sigmaos with kubernetes for orchestrating microservice and serverless applications*, M.Sc. Thesis

Teaching Experience

2023 **TA: Computer Systems Security**, *Massachusetts Institute of Technology*
Instructors: Nickolai Zeldovich

2022 **TA: Operating Systems Engineering**, *Massachusetts Institute of Technology*
Instructors: Frans Kaashoek, Robert Morris

2020 **TA: Concurrency (Honors)**, *The University of Texas at Austin*
Instructor: Christopher J. Rossbach

2020 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin

2019 **TA: Intro to CS Research (Honors)**, *The University of Texas at Austin*
Instructor: Calvin Lin

Service

2022-present **EECS Communication Lab Fellow**, *Massachusetts Institute of Technology*, Cambridge, MA

2021 **SOSP '21 Artifact Evaluation Committee**, *SOSP 2021*, Kolbenz, Germany

2020-present **MIT Graduate Application Assistance Program Mentor**, *Massachusetts Institute of Technology*, Cambridge, MA

Languages

English Fluent

Spanish Fluent

French Intermediate